Part - A

Ques1.
(C) 3 (2)

Ques2.
(C) Femur (2)

Ques3.
(A) Closing of Suture (2)

Ques4.
(A) Male Skull (2)

Ques5.
(A) Identifying Individuals. (2)

Ques6.
(C) Head Shape (2)

Ques7.

(D) Facial recoquition (1)

Ques8.

(D) All of the above (2)

Ques9.

(A) Age at death. (2)

Ques10.

(C) They are rarely altered. (1)

Part - B

Ques 11.1 (4)

Endochondral Ossification

- Ossification is also known as formation of bone.
- Osteoblasts are bone forming cells.
- ~~The~~ Ossification are of two types
  (i) Intrachondral
  (ii) Endochondral
- Endochondral ossification requires a cartilage model for ossification which is made up of hyaline.
- This is a ~~two~~ involves primary & secondary ossification.

→ Steps

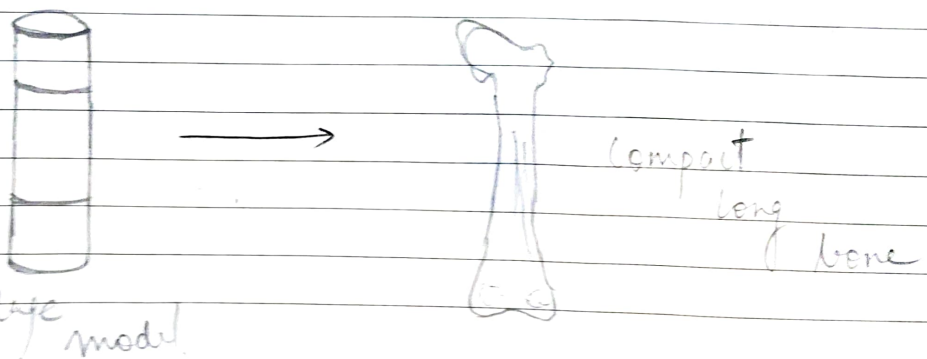
(1) Firstly cartilage model is formed. (Hyaline)

(2) Then this cartilage model formes into bone.

(3) Then, primary ossification takes place & develops into the epiphyses. (before birth)

(4) Then, secondary ossification takes place & forms the diphyses. (Takes place after birth)

(5) Then, finally the compact bone is formed.



Ques 12.1 (5)

Characteristics of Male Skull

1) Male skull is large as compare to female skull.
2) It is more robust in nature
3) It has square shaped orbits.
4) Eyebrows are more pronounced.
5) In male skull, the shaple of mandible is 'U'.
6) Jawlines are more prominent.
7) Ramus height is more as compared to female skull
8) The overall skull is very hard whereas female skull is smooth.

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9) In male skull, the forehead is slopy and broader.

10) Gonial angle is less in male skull.

11) Mandibular condyle is more prominent as compared to female skull.

Ques 13.1 (4.5)

1) Somatoscopy is the usual examination of a body in which physical features are observed for personal identification.

2) Darwin's Tubercle, earlobes, supra-orbital ridges and scar or occupational marks are forensically very important in personal identification.

3) Supra-orbital ridges are more prominent in males & less prominent in females.

4) Scar marks are the marks which are occured from an injury or self harm, they are very helpful in personal identification, they can be unique to the individuals.

5) The occupational marks are the marks which are occured because of the type of profession like :-

6) The site workers eat something & ~~many~~ & can have pigments on their teeths.

7) The bangles & iron factory workers may have burnt mark. & & many more.

8) Like this we can get to know about the occupation of individuals.

9) Similarly, Darwin's tubercle & ear lobes are also very imp in forensic for personal identification.

Ques14.1 (4)

b) The Portrait parle or the Bertillon system , these are given by Alphonse Bertillon. These are the type of facial reconstruction methods for the identification of individuals. In these methods the reconstruction of the parts like eyes, orbits, nose, lips, chin, ear are reconstructed on the basis of the knowledge of these methods.

- Where as, the modern biometric methods are very easy & requires only biometric for the identification of the individuals, which is less time taking & less complicated as compared to the portrait pasle / Bertillon system.

→ Portrait parle / Bertillon system are very
- complicated,
- requires more knowledge.
- takes more time
- more chances of less accurate.

→ Whereas, Biometric methods are digital,
- requires less time.
- more accurate.
- very simple & easy.

Ques 15.2 (4)

Facial reconstruction is a technique used for personal identification, used in anthropology.

There are many types in which facial reconstruction is possible like:-

(1) ~~Superimposition~~
The technique in which overlapping occurs. they are putted on one another & then analysing occurs for matching them

~~(2) Photograph~~

(2) 2D Reconstruction.
This is the method in which the photographical superimposition takes place & then examination for personal identity occurs.

(3) 3D Reconstruction

This is the method in which reconstruction takes place on a 3-D model & used for examination of personal identity.

(4) Portrait parle / Bertillon System.

(5) ~~Media Superimposition~~ Media Superimposition.

Manual	Digital.
- Requires skilled experts (sketch)	- No requirements of sketchers
- No digital tool requirements.	- Requires very costly tools.
- Takes lots of time	- Requires less time.
- Less accurate	- More accurate.

Part-C

Ques 16.1

Comparative skeleton anatomy of human & non-human bones:

Human	Non-Human
- Short skull	- skull is elongated.
- Chin is present.	- Mostly chin is absent
- Clavicle is more prominent.	- Clavicle is absent or less prominent.
- ~~Bid~~ Bipedal movement	- Quadripedal movement.
- Bone density is high	- Bone density is less.

Human	Non-Human
- Bones are inverted for ~~ted~~ bipedal movement	- Bones are straight for quarupedal movement.
- Small teeth.	- Bigger Canines.
- 'U' shaped skull.	- 'V' shaped skull.

Ques 20.1 (7)

1) Forensic odontology is basically the study related to teeth.

2) There is a very important role of forensic odontology in mass disaster situations.

3) In cases like mass disaster, or severing the teeths are the part of body which does not alter easily but that ~~it~~ time the the other evidences like ~~this~~ hair, fingerprints, DNA can be destroyed.

4) The teeths can be compared with the previous dental records ~~teheen~~ an can able to identify the victim.

5) With the helps of one's teeths we can able to determine the age of an individual; ~~Be~~ as ~~s~~ some teeths are erupt at ~~a~~ a

particular age only. for eg: Wisdom teeth ~ 21 to 25 years)

7) We can able to determine the sex of an individual with the help of teeth as female teeths are smaller in comparison to male.

8) Teeth are the only part of the body which can't be altered easily in the mass disaster unlike other evidence like fingerprints, etc.

9) We can able to get bite marks also & will examine & can links it with an individual.

10) INTERPOL consider dental (teeths) are the primary method for identification.

Ques 17.1 (15)

Male & Female skull

1) Male skull is large as compared to female skull.
2) In male skull the orbits are square in shape but in females they are oval in shape.
3) In male skull the supra-orbital ridges are more pronounced as compared to female.
4) Mandible of male skull is 'U' shape
5) The ramus height of male skull is more but in females the ramus height is less.
6) In male skull the gonial angle is less than female skull.

7) The texture of male skull is ragged & hard where as female skull have smooth texture.

~~Eyebrows are more prominent in males~~

8) Male skull is more robust.

9) Mandible is ~~shot~~ short of female skull but large of male skull.

10) Mandibular condyle is more prominent in male skull & less prominent in female skull.

Ques 18. 1

Somatometry

- Somato means body
- Metry means measurements

Somatometry means the measurements of the body

~~Head measurements~~

The measurements of different parts like head, face, nose, ear, height & body weight are taken with very much accuracy so that there will be no mistake in facial reconstruction and can identify individuals.

Somatometric measurements are taken with very accuracy so that the report must be clear & justice will be given to right one.

Geeta University, Panipat
21 MAY 2025
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These measurements are very reliable & needs experts with good knowledge for taking the measurements with 100% accuracy & give their report & opinion accordingly.

For personal identification this method is very important & accurate too in forensic examination across diverse population.

It is admissible in court too.

Ques 19.1 (7)

Facial superimposition is very useful in forensic science for determining the individuals.

There are many facial superimposition techniques like

- 2D (Photographic)
- 3D (Modeling)
- Video superimposition.
- Craniofacial superimposition.
- Portrait parle.
- Bertillon System.

→ Superimposition basically means overlapping (one on another).

→ They require very skilled & knowledge full expert

experts for the examination.

- This is very time taking process.
- This requires more accuracy
- Doesn't require much technology.

End of Writing